

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2024 ( Regular )**

**Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTEEC601 - SWITCHGEAR AND PROTECTION**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks
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3. Question No. 1 will be compulsory and includes objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

The minimum current at which a relay starts operating is called:

- a) Plug setting
- b) Pick-up current
- c) Reset current
- d) Time setting

Protection zones are arranged to:

- a) Leave parts unprotected
- b) Overlap adjacent zones
- c) Reduce relay sensitivity
- d) Increase generation

Differential relays operate on the principle of backup protection.

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- a) Voltage comparison
- b) Current difference comparison
- c) Frequency comparison
- d) Power factor comparison

an principle of attracted armature type relay.

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**Q2. Solve**

- A) Discuss the working principle of microprocessor-based directional relays.
- B) Compare static ratings of numerical relays.

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**Q4. Solve Any Two of the following.**

- A) Explain current chopping & its effects and methods to reduce it.
- B) Discuss the construction, operation, and advantages of SF6 circuit breakers.
- C) Explain various ratings of circuit breakers.

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**Q5. Solve Any Two of the following.**

- A) Discuss the need and importance of distance protection.
- B) Explain the protection scheme used for ring main feeders.
- C) Compare MCBs and fuses.

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**Q6. Solve Any Two of the following.**

- A) Explain the differential protection scheme used for alternators.
- B) Explain why backup protection is necessary in power systems.
- C) Explain the role of IS/IEC standards in switchgear and protection equipment selection.

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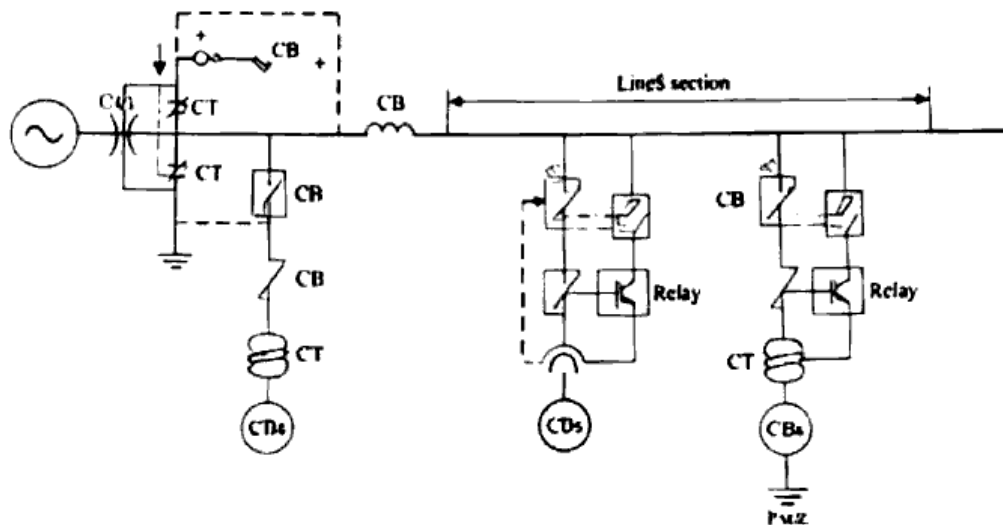
Q2. Solve the following.

- A) Differentiate the temporary primary protection and backup protection. 6
- B) Describe the construction and working advantages of SF6 circuit breakers type relay. 6

Q3. Solve the following.

- A) Discuss the working principle of microprocessor-based directional relays. 6
- B) Explain the protection scheme used for ring main feeders. 6
- C) Compare MCBs and fuses.

Q4. Directional overcurrent protection scheme used for radial feeder systems in direction that schematic diagram for radial scheme directional over system/in scheme for radial feeder system.



- A) Explain the differential primary starting principle of attracted armature type relay. 6

Q2. Solve the following.

- A) Discuss the working principle of microprocessor-based directional relays. 6
- B) Compare static ratings of numerical relays. 6

Q4. Solve Any Two of the following.

- A) Explain current chopping & its effects and methods to reduce it. 6
- B) Discuss the construction, operation, and advantages of SF6 circuit breakers. 6
- C) Explain various ratings of circuit breakers. 6

Q5. Solve Any Two of the following.

- A) Discuss the need and importance of distance protection. 6
- B) Explain the protection scheme used for ring main feeders. 6
- C) Compare MCBs and fuses. 6

Q5. Solve Any Two of the following.

- A) Explain the differential protection scheme used for alternators. 6
- B) Explain why backup protection is necessary in power systems. 6
- C) Explain the role of IS/IEC standards in switchgear and protection equipment selection. 6